

Indian Statistical Institute
M.Tech. (CS), First Year, Final of Second Semester Examination, 2018-19
Database Management Systems

Full Marks: 100

Date: 03-05-2019

Time: 3 Hours

Answer any *ten* of the following questions

$10 \times 10 = 100$

1. For any arbitrary relations R_1, R_2, R_3, R_4 , where $R_2 \supseteq R_1$ and $R_4 \supseteq R_3$, prove or disprove the following:

(a) $R_2 \times R_4 \supseteq R_1 \times R_3$.

(b) $R_2 - R_4 \supseteq R_1 - R_3$.

5+5

2. (a) Express the division operation in relational algebra in terms of Cartesian product and other operations. Justify your answer.

(b) Consider a schema with the following three relations, with the primary keys underlined therein, as follows.

REL1 = $\langle \underline{\text{attribute1}}$: integer, attribute2 : string $\rangle$

REL2 = $\langle \underline{\text{attribute3}}$: integer, attribute4 : string $\rangle$

REL3 = $\langle \underline{\text{attribute1}}$: integer, $\underline{\text{attribute3}}$: integer $\rangle$

Decompose the following query into single-variable queries.

select REL1.attribute2 from REL1, REL2, REL3 where REL1.attribute1 = REL3.attribute1 and REL2.attribute3 = REL3.attribute3 and REL2.attribute4 = "DBMS";

(2+3)+5

3. Consider the following schema of an online code repository system.

CONTRIBUTOR = $\langle \underline{\text{contributor_id}}$: integer, contributor_name : string $\rangle$

CODE-GROUP = $\langle \underline{\text{contributor_id}}$: integer, $\underline{\text{code_group}}$: string, num_submit : integer $\rangle$

The keys corresponding to each relation are underlined. Write the following queries in SQL.

(a) Find all the contributors who have made at most one submission within the R code group.

(b) Find all the contributors who have made at least two submissions within the Python code group.

5+5

4. Consider a relational schema $R = \langle \text{STUVWXYZ} \rangle$, where all the fields in R contain atomic values. Let $F = \{UZ \rightarrow Y, S \rightarrow TU, T \rightarrow UXZ, W \rightarrow S, X \rightarrow WY\}$ be a set of functional dependencies such that F^+ is exactly the set of functional dependencies that hold for R .

- (a) Find out the candidate keys of R .
 (b) What is the highest possible normal form that R satisfies? Justify your answer. 5+(1+4)
5. (a) Justify the statement “A *conflict serializable* schedule is always *view serializable* but not the vice versa.”
 (b) Let there be a relation with data items A and B . Derive the *precedence graph* from the following schedule of transactions working on the said relation. Conclude whether the given schedule is *conflict serializable* or not.

Transaction T_1	Transaction T_2	Transaction T_3
read(A)		
	read(B)	
	write(A)	
	Commit	
write(A)		
Commit		
		write(A)
		Commit

- 3+(6+1)
6. (a) Explain the concept of *uncommitted dependency* (or *dirty read*), which might happen while applying a two-phase locking protocol on databases aiming concurrency control.
 (b) What is phantom phenomenon? How can it be solved during concurrency control? 4+(2+4)
7. Let there be a relation with data items A and B having initial values 0 and 100, respectively. A *log-based recovery* scheme is used with *immediate database modification* to keep control over the updates in this relation. A pair of transactions T_1 and T_2 works on this relation as follows.

Transaction	Instruction
T_1	read(A)
T_1	$A \leftarrow A + 20$
T_1	read(B)
T_1	$B \leftarrow A$
T_1	write(A)
T_1	write(B)
T_2	read(A)
T_2	$A \leftarrow A * 2$
T_2	write(A)

Write down the entries to the log record file and recovery steps to be carried out against each instruction shown above.

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8. Consider a distributed dynamic environment with 3 nodes and 25 data objects. The nodes and objects are labeled as N_i ($i = 100, 200, 300$) and O_i ($i = 0, 1, \dots, 24$), respectively. A circular array C (with index 0, 1, ... 49) is used to implement *consistent hashing* for load balancing among the said nodes while processing the data objects. The node N_i is hashed at

$C[i\%80]$ and the object O_i is hashed at $C[2i+1]$. Let there be the following data objects processed on those nodes at a given instance of time.

$O_{11}, O_2, O_{15}, O_7, O_3, O_{19}, O_8, O_{13}$

Suppose an existing node departs and a new node N_{400} comes in with no effect on the current loads. Then identify the node that departed with appropriate justifications.

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9. Suppose a company is willing to develop a mobile based application, say GREET. The purpose of this application is to study public reactions during musical reality shows (like Indian Idol). The account holders on GREET can only post audios and can only reply through Emojis (e.g., ☺). Perform a case study on the database systems required to support this. You are expected to suggest the data model, schema, and storage implementation details, with appropriate justifications.

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10. Give appropriate justifications of the following things:

- (a) Settling transaction logs and temporary spaces properly are crucial for database tuning.
- (b) Thomas' write rule for timestamp-based protocols reduces the amount of transaction rollbacks.

5+5

11. Give appropriate examples of the following things:

- (a) Total and partial participation of entity sets in an entity-relationship model.
- (b) System R style optimization of queries.

5+5
